

Gaussian auxiliaries, up to time sharing, are optimal for the rate–leakage region with decoder side information and a weighted distortion

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This note extends the companion manuscript *Rate, Leakage, and Distortion for Gaussian Sources* (tit-rate-leakage.tex, Section on decoder side information and Proposition decbounds) and uses its notation. Its only outside ingredient is the single-letter extremal inequality of Xu and Chen [1, Theorem 4, eq. (200)], together with the chain-rule identities of their eqs. (196)–(199). Their operational characterization of the matrix-distortion region is not used, because its converse rests on a discrete-alphabet single-letterization [1, eq. (196), citing Villard and Piantanida]. The continuous-source single-letterization is taken from the manuscript’s Theorem decregion instead. The equation numbers are those of the preprint. The published version [2] describes its proofs as simplified and corrected and has not been checked against the form used here.

1 Setting and the Gaussian program

A jointly Gaussian source $T \sim \mathcal{N}(0, \Sigma_T)$, $\Sigma_T \succ 0$, is described by a deterministic encoder $M = f_n(T^n)$. The decoder sees (M, C^n) and must reproduce $Y = FT$ under the weighted distortion with weight $Q = F^\top W F$, $W \succ 0$. The holder sees S^n . The side informations are linear Gaussian observations of T with information matrices $J_C, J_S \succeq 0$, and C and S are conditionally independent given T . Rate, distortion, and leakage $L_n = \frac{1}{n} H(M | S^n)$ are as in the manuscript, and $\mathcal{A}(D)$ is the closure of the set of pairs (R, L) achievable at distortion D . Write $\Sigma_{T|C} = (\Sigma_T^{-1} + J_C)^{-1}$ and let $D > 0$.

For matrices $0 \prec B \preceq A \preceq \Sigma_T$ define, in bits,

$$\begin{aligned} r(A, B) &= \frac{1}{2} \log \det(\Sigma_{T|C}(B^{-1} + J_C)), \\ \ell(A, B) &= r(A, B) + \frac{1}{2} \log \frac{\det(I + J_C \Sigma_T)}{\det(I + J_C A)} - \frac{1}{2} \log \frac{\det(I + J_S \Sigma_T)}{\det(I + J_S A)}, \\ \Lambda(B) &= (B^{-1} + J_C)^{-1} = B(I + J_C B)^{-1}. \end{aligned}$$

For a jointly Gaussian pair $V_0 \subseteq V$ (that is, V_0 a sub-vector of V) with $V = T = (C, S)$, $A = \text{Cov}(T | V_0)$, and $B = \text{Cov}(T | V)$, these are $I(T; V | C)$, $I(T; V | C) + I(V_0; C) - I(V_0; S)$, and $\text{Cov}(T | V, C)$, because $I(V_0; C) = I(T; C) - I(T; C | V_0)$ and similarly for S . The *Gaussian region* is

$$\mathcal{G}(D) = \overline{\text{conv}}\{(r(A, B), \ell(A, B)) : 0 \prec B \preceq A \preceq \Sigma_T, \text{tr}(Q\Lambda(B)) \leq D\} + \mathbb{R}_+^2,$$

and for a matrix $\Lambda_0 \succeq 0$ the matrix-distortion Gaussian region $\mathcal{G}_m(\Lambda_0)$ is defined the same way with the constraint $\Lambda(B) \preceq \Lambda_0$ (it is empty when Λ_0 is singular). Both are closed, because the closed convex hull is bounded below ($r \geq 0$ and $\ell \geq 0$) and adding $\mathbb{R}_+^2$ to a closed set bounded below keeps it closed. They are convex and closed upward in (R, L) .

Theorem 1. For all $J_C, J_S \succeq 0$ and all $D > 0$, $\mathcal{A}(D) = \mathcal{G}(D)$.

2 Proof

Lemma 1 (Inner bound). $\mathcal{G}(D) \subseteq \mathcal{A}(D)$.

Proof. Given $0 \prec B \preceq A \preceq \Sigma_T$, let $V_0 = G_0 T + N_0$ with $G_0^\top G_0 = A^{-1} - \Sigma_T^{-1}$ and unit noise, so $\text{Cov}(T | V_0) = A$, and let $V = (V_0, G_1 T + N_1)$ with $G_1^\top G_1 = B^{-1} - A^{-1} \succeq 0$ and an independent unit noise, so $\text{Cov}(T | V) = B$. All noises are independent of (T, C, S) , so $V_0 - V - T - (C, S)$ and $I(T; V) < \infty$. Proposition decbounds(a) of the manuscript then makes (r, ℓ) achievable at distortion $\text{tr}(Q\Lambda(B))$. Pairs above an achievable pair are achievable, time sharing gives convex combinations, and $\mathcal{A}(D)$ is closed by definition. $\square$

Lemma 2 (Continuity). Let the parameters $p = (J_C, J_S, D)$ converge, $p_k \rightarrow p$, and let $(R_k, L_k) \in \mathcal{G}(p_k)$ with $(R_k, L_k) \rightarrow (R, L)$. Then $(R, L) \in \mathcal{G}(p)$.

Proof. On the compact set $\Theta = \{0 \preceq B \preceq A \preceq \Sigma_T\}$, the map $(A, B) \mapsto \Lambda(B) = B(I + J_C B)^{-1}$ is continuous jointly with J_C , so the constraint set is closed. The two log-determinant ratios in ℓ are continuous on Θ jointly with (J_C, J_S) and bounded in absolute value by $\frac{1}{2} \log \det(I + J_C \Sigma_T) + \frac{1}{2} \log \det(I + J_S \Sigma_T)$. Writing $r = \frac{1}{2} \log \det \Sigma_{T|C} + \frac{1}{2} \log \det(I + J_C B) - \frac{1}{2} \log \det B$ shows that r , extended by $+\infty$ where $\det B = 0$, is lower semicontinuous on Θ jointly with J_C , and so is ℓ . Both are bounded below by a constant $-c$ uniform over a neighbourhood of p , since $r \geq 0$ and the ratios are bounded.

By Carathéodory's theorem in $\mathbb{R}^2$ and the definition of the closure, for each k there are (A_j^k, B_j^k) feasible for p_k and weights $w_j^k \geq 0$ summing to one, $j = 1, 2, 3$, with $\sum_j w_j^k(r, \ell)(A_j^k, B_j^k; p_k) \leq (R_k, L_k) + (1/k, 1/k)$ componentwise. Pass to a subsequence along which $(A_j^k, B_j^k) \rightarrow (\bar{A}_j, \bar{B}_j) \in \Theta$ and $w_j^k \rightarrow \bar{w}_j$. Each limit point is feasible for p by closedness. For j with $\bar{w}_j > 0$, lower semicontinuity gives $(r, \ell)(\bar{A}_j, \bar{B}_j; p) \leq \liminf (r, \ell)(A_j^k, B_j^k; p_k)$, and the terms with $\bar{w}_j = 0$ contribute at least $-cw_j^k \rightarrow 0$. Hence $\sum_j \bar{w}_j(r, \ell)(\bar{A}_j, \bar{B}_j; p) \leq (R, L)$, which forces $\det \bar{B}_j > 0$ whenever $\bar{w}_j > 0$, and $(R, L) \in \mathcal{G}(p)$. $\square$

Lemma 3 (Monotonicity). If $J_C \preceq J'_C$ and $J_S \preceq J'_S$, then $\mathcal{A}(D; J_C, J_S) \subseteq \mathcal{A}(D; J'_C, J'_S)$.

Proof. This is the argument of Proposition decbounds(b) and (c) of the manuscript, applied in turn. A decoder that sees an observation with information J'_C can simulate one with information J_C , after which derandomization leaves a deterministic decoder with no larger distortion and the same leakage. A holder with information J'_S can simulate one with information J_S , so $H(M | S'^n) \leq H(M | S^n)$ for every code. $\square$

Lemma 4 (Single-letter extremal bound). Let $J_C \succ 0$, $J_S \succ 0$, $\Lambda_0 \succ 0$, and $\mu \geq 0$. For every $V_0 - V - T - (C, S)$ with $I(T; V) < \infty$ and $\mathbb{E} \text{Cov}(T | V, C) \preceq \Lambda_0$,

$$\mu I(T; V | C) + [I(T; V | C) + I(V_0; C) - I(V_0; S)] \geq \min \{ \mu r(A, B) + \ell(A, B) : 0 \prec B \preceq A \preceq \Sigma_T, \Lambda(B) \preceq \Lambda_0 \}.$$

Proof. With $J_C \succ 0$ the statistic $Y = J_C^{-1} H_C^\top \Sigma_{N_C}^{-1} C$ is sufficient for T in C and equals $T + N_Y$ with $\text{Cov } N_Y = J_C^{-1}$, and likewise S reduces to $Z = T + N_Z$ with $\text{Cov } N_Z = J_S^{-1}$. Since Y is a sufficient statistic computed from C , and $V_0 - V - T - (C, S)$ with C and S conditionally independent given T , the chain $C - Y - (T, V, V_0, S)$ holds. Hence $I(T; V | C) = I(T; V | Y)$, $I(V_0; C) = I(V_0; Y)$, and $\mathbb{E}[T | V, C] = \mathbb{E}[T | V, Y]$, so every term on the left and $\mathbb{E} \text{Cov}(T | V, C)$ are unchanged when

C is replaced by Y , and likewise for S and Z . This is the setting of [1, Sec. V] with $X = T$, $K = \Sigma_T$, $K_Y = J_C^{-1}$, $K_Z = J_S^{-1}$, $U = V_0$, and distortion matrix Λ_0 . By the chain rule, as in their eqs. (196)–(199), the left side equals

$$-(\mu + 1)h(T | Y, V) + h(T | Y, V_0) - h(T | Z, V_0) + c_\mu,$$

where $c_\mu = \mu h(T | Y) + h(T | Z)$ and all entropies are in bits. The program (189) has a minimizer. Its objective is at least $r \rightarrow \infty$ as $\det B \rightarrow 0$, because $\ell - r$ is bounded, so the minimum may be taken over a compact set $\{B \succeq \delta I\}$. Its constraints are linear matrix inequalities with a strictly feasible point ($B_1 = B_2 = tI$ for large t), so a minimizer (B_1^*, B_2^*) satisfies the KKT conditions (191)–(195) required by their Theorem 4, and $\Lambda_0 \succ 0$ as that theorem requires. Their extremal inequality [1, Theorem 4, eq. (200)] then bounds the expression above from below by its Gaussian value at (B_1^*, B_2^*) . The substitution $A^{-1} = \Sigma_T^{-1} + B_2$, $B^{-1} = \Sigma_T^{-1} + B_1 + B_2$ maps $\{B_1, B_2 \succeq 0, B_1 + B_2 \succeq \Lambda_0^{-1} - \Sigma_T^{-1} - J_C\}$ bijectively onto $\{0 \prec B \preceq A \preceq \Sigma_T, \Lambda(B) \preceq \Lambda_0\}$, and it maps the right side of (200), as a function of (B_1, B_2) , plus c_μ , which is the objective of (189) plus $h(T | Z)$ because (189) already contains the term $\mu h(T | Y)$, to $\mu r(A, B) + \ell(A, B)$, because $\log \det(I + J_C A) = \log \det A + \log \det(A^{-1} + J_C)$ and similarly for J_S , so the terms in $\log \det A$ cancel. $\square$

Proof of Theorem 1. Lemma 1 gives $\mathcal{G}(D) \subseteq \mathcal{A}(D)$.

Full rank. Let $J_C, J_S \succ 0$ and let (R, L) be achievable at D . By the manuscript's Theorem decretion, for every $\epsilon > 0$ there are $V_0 - V - T - (C, S)$ with $I(T; V) < \infty$ and a reconstruction from (V, C) with distortion at most $D + \epsilon$, such that $R + \epsilon \geq I(T; V | C)$ and $L + \epsilon \geq I(T; V | C) + I(V_0; C) - I(V_0; S)$. Let $\Lambda_0 = \mathbb{E} \text{Cov}(T | V, C)$. The distortion is at least $\text{tr}(Q\Lambda_0)$, so $\text{tr}(Q\Lambda_0) \leq D + \epsilon$, and $\Lambda_0 \succ 0$ because $I(T; V | C) < \infty$ (proof of the manuscript's Theorem decsi). By Lemma 4, $\mu(R + \epsilon) + (L + \epsilon)$ is at least the least value of $\mu R' + L'$ over $\mathcal{G}_m(\Lambda_0)$ for every $\mu \geq 0$. Dividing by μ and letting $\mu \rightarrow \infty$ gives the same for the direction $(1, 0)$. Every other direction has infimum $-\infty$ over both $\{(R + \epsilon, L + \epsilon)\} + \mathbb{R}_+^2$ and $\mathcal{G}_m(\Lambda_0)$, and a closed convex set that is closed upward is the intersection of its supporting half-planes in the directions $(\mu, 1)$, $\mu \geq 0$, and $(1, 0)$. Hence $(R + \epsilon, L + \epsilon) \in \mathcal{G}_m(\Lambda_0) \subseteq \mathcal{G}(D + \epsilon)$, the inclusion because $Q \succeq 0$ gives $\text{tr}(Q\Lambda(B)) \leq \text{tr}(Q\Lambda_0) \leq D + \epsilon$. Lemma 2 with $(R + \epsilon, L + \epsilon) \rightarrow (R, L)$ as $\epsilon \rightarrow 0$ gives $(R, L) \in \mathcal{G}(D)$.

General J_C, J_S . For $\epsilon > 0$, Lemma 3 and the full-rank case give $\mathcal{A}(D; J_C, J_S) \subseteq \mathcal{A}(D; J_C + \epsilon I, J_S + \epsilon I) = \mathcal{G}(D; J_C + \epsilon I, J_S + \epsilon I)$. Letting $\epsilon \rightarrow 0$, Lemma 2 gives $\mathcal{A}(D) \subseteq \mathcal{G}(D)$. When $D \geq \text{tr}(Q\Sigma_{T|C})$, the choice $A = B = \Sigma_T$ gives the pair $(0, 0)$ and both regions are $\mathbb{R}_+^2$. $\square$

Remark 1. The theorem says that the second auxiliary of the manuscript's Theorem decretion can be taken jointly Gaussian and nested in the description, up to time sharing, for every pair of linear Gaussian observations and every weighted quadratic distortion. Every exposed boundary point of $\mathcal{A}(D)$ is attained by a single Gaussian pair (A, B) , since the minimum of $\mu r + \ell$ over the generating set is attained, and the rest of the boundary by time sharing between two such pairs. It reduces the open region of Proposition decbounds to a finite-dimensional program over $0 \prec B \preceq A \preceq \Sigma_T$. That program is not convex in general, and its explicit solution, an analogue of the manuscript's tilted water-filling with two auxiliaries, is not given here. In the degraded cases the optimum can be taken with $A = \Sigma_T$ (V_0 constant) or $A = B$ ($V_0 = V$), which recovers Theorem decsi. The weighted-to-matrix reduction is routine. What is new is the combination of the extremal inequality with the continuous-source single-letterization of the manuscript, and the extension to singular J_C and J_S by enhancement and continuity. The weight needs no assumption beyond $Q \succeq 0$.

References

- [1] Y. Xu and G. Chen, “New proofs of extremal inequalities with applications,” arXiv:2109.00377, 2021.
- [2] Y. Xu, G. Chen, J. Chen, and S. Jin, “New proofs of Gaussian extremal inequalities with applications,” *IEEE Trans. Inf. Theory*, vol. 70, no. 5, pp. 3082–3099, May 2024.